

AI Self-preferencing in Algorithmic Hiring: Empirical Evidence and Insights

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Problem definition: As artificial intelligence (AI) tools become widely adopted, large language models (LLMs) are increasingly involved on both sides of decision-making processes, ranging from hiring to content moderation. This dual adoption raises a critical question: do LLMs systematically favor content that resembles their own outputs? Prior research in computer science has identified self-preference bias, i.e., the tendency of LLMs to favor their own generated content, but its real-world implications have not been empirically evaluated. **Methodology/results:** We examine this question in the context of algorithmic hiring, a high-volume screening process central to workplace operations. In this setting, job applicants increasingly rely on LLMs to refine resumes, while employers deploy similar tools to screen those same materials. Using a large-scale controlled resume correspondence experiment, we find that LLMs consistently prefer resumes generated by themselves over those written by humans or produced by alternative models, even when content quality is controlled. The bias against human-written resumes is particularly substantial, with self-preference bias ranging from 67% to 82% across major commercial and open-source models. **Managerial implications:** To quantify the operational impact of self-preference bias, we simulate realistic hiring pipelines across 24 occupations. These simulations show that candidates using the same LLM as the evaluator are 23% to 60% more likely to be shortlisted than equally qualified applicants submitting human-written resumes, with the largest disadvantages observed in business-related fields such as sales and accounting. We further demonstrate that, in many cases, this bias can be reduced by more than 50% through simple interventions that target LLMs’ self-recognition capabilities. These findings highlight an emerging but previously overlooked risk in AI-assisted decision making and call for expanded frameworks of AI fairness in business operations that address biases from AI-AI interactions.

Key words: generative AI, algorithmic hiring, self-preference, future of work, AI fairness, empirical study

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1. Introduction

The rapid development and commercialization of artificial intelligence (AI) have made large language models (LLMs) widely accessible across both professional and everyday contexts. As these

tools become embedded in diverse workflows, they are increasingly involved on both sides of content generation and evaluation. In hiring, for instance, applicants often use LLMs to draft or refine their resumes, while employers leverage similar tools to screen or rank candidates (New York State Society of CPAs 2024, ResumeBuilder.com 2023, Wiles et al. 2025). On social media, users frequently use LLMs to help compose or polish posts, while platforms may employ LLMs to moderate content, e.g., flagging, categorizing, or filtering user submissions (e.g., Kumar et al. 2024). In academia, researchers may use LLMs to improve their manuscripts, while conferences and journals are beginning to experiment with LLM-assisted peer review (e.g., Thakkar et al. 2025). Similar patterns are emerging in education and customer service, where LLMs support both communication and assessment tasks (Forbes 2024). In these domains and beyond, LLMs increasingly *both* generate and evaluate the same content, giving rise to a new class of AI–AI interactions with significant implications for human decision-making and business operations (Laurito et al. 2025).

Recent research in computer science has identified a behavioral tendency in LLMs known as self-preference, i.e., the inclination of a model to favor content it generated itself over that written by humans or produced by alternative models (e.g., Panickssery et al. 2024). While this phenomenon has been documented in benchmark evaluations (e.g., Zheng et al. 2023, Bai et al. 2023), its implications for real-world decision-making remain largely unexplored. As LLMs are increasingly deployed in high-stakes settings where they may evaluate content also generated by LLMs, self-preference introduces a novel form of bias. Unlike traditional biases rooted in demographic disparities (e.g., Sheng et al. 2019), this bias emerges endogenously from AI–AI interactions, in which the model’s own evaluative behavior systematically favors outputs aligned with its generative patterns.

Self-preferencing bias is inherently interactional: it arises when LLMs are asked to judge content that may share stylistic or linguistic patterns with their own generative outputs. As such, it poses a new challenge for AI fairness, one that is not addressed by existing safeguards focused on demographic disparities. If left unchecked, self-preference could subtly distort evaluative processes across hiring, education, publishing, and more—privileging those who employ the same AI system used for evaluation (i.e., the “right” tool from the model’s perspective) while disadvantaging those who use different tools or none at all. Addressing this issue will require expanding current fairness frameworks to account for LLMs’ dual roles as both decision aids and evaluators in an increasingly AI-mediated business environment.

Among the many domains where self-preference bias may arise, algorithmic hiring is particularly consequential. Employers are now routinely adopting LLMs to streamline resume screening and candidate ranking, often as part of automated workflows that support human decision-making (e.g., Gan et al. 2024, Kim et al. 2024, ResumeBuilder 2024, Sarumathi et al. 2025). Unlike traditional keyword-matching systems, LLMs can evaluate resumes in a more holistic manner—synthesizing

content, inferring intent, and making contextual judgments beyond simple heuristics (Pritchett 2025). While this shift promises greater efficiency and scalability, it may also magnify the risk of bias if an LLM systematically favors resumes that reflect its own generative style. In such cases, evaluations may hinge less on the substantive quality of a candidate’s credentials and more by superficial stylistic alignment with the evaluator LLM, conferring unwarranted advantages on applicants who use the same model to compose their materials. In effect, this bias rewards access to specific generative technologies and penalizes those without it, even when applicants are otherwise equally qualified.

In this paper, we provide the first empirical evidence that self-preference bias can distort candidate evaluations in algorithmic hiring. Specifically, we examine whether LLMs, when deployed as evaluators, systematically favor resumes they generated themselves over otherwise equivalent resumes written by humans or produced by alternative LLM models. To test this, we conduct a large-scale resume correspondence experiment using a real-world dataset of 2,245 human-written resumes, sourced from a professional resume-building platform prior to the widespread adoption of generative AI. For each resume, we generate multiple counterfactual versions using a range of state-of-the-art LLMs, including GPT-4o, GPT-4o-mini, GPT-4-turbo, LLaMA 3.3-70B, Mistral-7B, Qwen 2.5-72B, and Deepseek-V3.¹ Having content quality controlled, we assess whether these LLMs exhibit systematic bias in favor of their own outputs when acting as evaluators.

We distinguish between two forms of self-preference bias: *LLM-vs-Human*, where a model prefers its own generated content over a human-written equivalent; and *LLM-vs-LLM*, where a model favors its own output over content produced by an alternative (different) LLM. We find strong and consistent evidence of LLM-vs-Human self-preferencing across most models. Larger models, such as GPT-4o, GPT-4-turbo, DeepSeek-V3, Qwen-2.5-72B, and LLaMA 3.3-70B, exhibit particularly strong bias, exceeding 65% even after controlling for content quality and reaching over 80% for GPT-4o. By contrast, LLM-vs-LLM self-preferencing is more heterogeneous. DeepSeek-V3 shows the strongest bias in this setting, preferring its own outputs by 69% against LLaMA 3.3-70B and 28% against GPT-4o. GPT-4o and LLaMA 3.3-70B, in comparison, do not display consistent preferences when evaluating content generated by other models.

To assess the labor market implications of this bias, we simulate hiring pipelines across 24 occupations. In these simulations, candidates using the same LLM as the evaluator are about 23–60% more likely to be shortlisted than equally qualified applicants submitting human-written

¹ These models represent the state of the art among widely deployed LLMs at the time of our study. Although newer and more capable models will continue to emerge, the key pattern we examine—LLMs being used both to generate content and to evaluate that content—already appears in many application domains and is expected to remain common. Our focus is therefore on a structural feature of AI-mediated decision making rather than on the behavior of any specific model version.